# GenAI for Engineers Learner Guide

*IES | v2.0 | One-day Copilot in Excel workshop*

IES | Original synthetic CW-401 engineering case. Values and limits are fictional teaching assumptions, not plant operating instructions. Generated references are independently calculated examples, not recorded Copilot executions.

Three connected labs replace the earlier16-lab programme, following the latest user direction. Use one realistic synthetic case to produce an SPC analysis, FMEA workbook and DMAIC report. No agent-building or SharePoint-administration labs are required.

## Before class

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged.

## RISE for Excel

Role: engineering data reviewer. Input: named tables and source revision. Steps: audit, calculate, chart and verify. Expectations: workbook output, source links, explicit limitations. E-MAP adds CTQ, metrics, assumptions and permitted scope.

## Good constraints

Keep RawTests unchanged; separate baseline from monitoring; preserve units and missingness; show formulas; do not invent cause or approval. Ask for missing information when it changes the result.

## Multiple personas and debate

Use reliability, operations and safety as simulated review lenses for the FMEA. Ask for separate positions, one evidence-linked challenge each, then a moderator summary that preserves disagreements. These are not independent experts.

## Benefits and evaluation

Clear prompts can improve consistency and reduce avoidable rework. Measure the quality of formulas, source fidelity and total review-inclusive time. Long prompts or persona agreement do not guarantee correctness.

# The CW-401 case and evidence

industrial cooling-water utility uses duty/standby metering pumps DP-401A/B. The workshop CTQ is measured delivery during a standardised collection test at fixed commanded setting. The design-note teaching target is 62.0 L/h with an acceptance band of 58.9–65.1 L/h; this is a delivery-test specification, not a chemical dose or water-quality compliance limit. Chemical identity and operating procedures are deliberately outside the case.

A repeatability study and comparison of pump delivery are needed before a maintenance recommendation. The team has 60 baseline collection tests, 30 subsequent monitoring tests and 30 post-intervention pilot tests. Collection volume is measured over 120 seconds; delivery L/h = collected litres × 30. The command setting is fixed for comparability; shift and pump identity are retained to detect confounding.

The monitoring period includes a sustained delivery reduction coinciding with a differential-pressure increase. A work order records inspection and a strainer replacement before the pilot. This temporal association does not isolate root cause: the pilot also follows a verification check, so intervention effects are confounded. No randomised causal claim is justified.

Raw export 121 rows; 120 unique tests; 118 comparable tests. WeeklyEvents is a separate eight-week event log with its own opportunities. FMEA scores are provisional workshop judgements. See ScoringNote before rating.

## One-day timetable

| Time | Block | Delivery focus |
|---|---|---|
| 09:00–09:30 | Copilot setup and prompt framework | 30 min |
| 09:30–11:15 | Lab01 SPC: quality audit, formulas and baseline | 105 min |
| 11:15–11:30 | Tea | Break |
| 11:30–12:05 | Lab01 SPC: chart and interpretation | 35 min |
| 12:05–13:00 | Lab02 FMEA: structure and ratings | 55 min |
| 13:00–14:00 | Lunch | Break |
| 14:00–14:45 | Lab02 FMEA: prioritisation and review | 45 min |
| 14:45–15:15 | Lab03 DMAIC: define and measure | 30 min |
| 15:15–15:30 | Tea | Break |
| 15:30–16:30 | Lab03 DMAIC: analyse, improve and control | 60 min |
| 16:30–17:00 | Workbook presentations and review | 30 min |

# Lab 01 Copilot in Excel: data quality and SPC

140 minutes | Lab-01-SPC

## 1. Prepare the workbook

Save a working copy of Starter.xlsx to your approved OneDrive or SharePoint location. Open it in Excel with your work/school account and use the Copilot side pane. Select the named table before prompting. If Copilot cannot edit, request the formula and insert it manually; record the fallback. Confirm tenant entitlement and supported workbook format before class. Keep the local raw files unchanged. Open ReadMe, Parameters, RawTests and Prompts. The workbook starts with source tables and blank output sheets.

## 2. Audit without deleting evidence

Select RawTests and submit prompt SPC1 from Prompts. Inspect TEST-068 (missing volume), TEST-083 (110-second invalid test) and duplicate TEST-088. Keep one copy of the duplicate; exclude the two invalid tests from the comparable analysis. Retain121 raw rows,120 unique tests and118 valid tests. Write decisions into QualityAudit.

## 3. Build the analysis table

Ask Copilot to filter and sort the valid unique records into Analysis. ValidTests is the prepared comparison source if cleaning stalls. Keep test_id, timestamp, phase, shift, pump, collected_L and differential pressure. Add delivery_L_h using collected_L*3600/Parameters duration. Spot-check one baseline and one monitoring row with a calculator.

## 4. Calculate a fixed baseline

Submit SPC2. Only60 baseline records estimate the centre and moving ranges. First baseline MR is blank; next59 are absolute successive differences. Mean62.00250 L/h; MRbar0.73881; sigma0.65498; LCL60.03757; UCL63.96743. Compare formulas with the reference after attempting the task.

## 5. Chart process behaviour

Submit SPC3. Use time order and phase labels; add constant mean, UCL and LCL series. Keep baseline limits for monitoring and pilot. There are28 valid monitoring and30 pilot tests. Seventeen monitoring points TEST-073–090 except083 are outside the limits. The missing record and invalid duration remain in the quality ledger.

## 6. Discuss capability and evidence

Submit SPC4. Conditional baseline Cp1.57766 and Cpk1.57639 use supplied specs. Check baseline behaviour and measurement assumptions before interpreting them. Do not transfer that conclusion to the shifted monitoring period. Note the differential-pressure association and the two changes before pilot; neither establishes a sole cause.

## 7. Retain the result

Save your workbook, prompts, chart, quality decisions, checks and a100-word interpretation. Record what Copilot actually did and what you calculated manually. Use the lab evaluation checklist and compare with the worked reference only after your attempt.

# SPC Copilot prompts

## SPC1 Audit the source

Using RawTests, audit test_id uniqueness, missing collected_L, duration_s and quality_flag. Do not modify RawTests. Create a QualityAudit table with record ID, issue, inclusion decision and reason. Use one unique record per test, require VALID and 120 seconds, and never discard an excursion solely because it is extreme. Report raw, unique and valid counts.

## SPC2 Calculate the baseline

Use ValidTests in time order. In Analysis create delivery_L_h = collected_L*3600/120. Use only the 60 baseline records to calculate mean and successive moving ranges; do not bridge across phases. Estimate sigma = MRbar/1.128 and individuals limits mean +/-3*sigma. Put constants and specifications in Parameters and show formulas, units and referenced rows.

## SPC3 Plot and interpret

Create an Excel line chart of delivery_L_h by timestamp for baseline, monitoring and pilot, with the fixed baseline mean/UCL/LCL. Flag points outside the limits without refitting them. Report phase counts and affected IDs. Explain the monitoring shift as a signal requiring investigation, not a proven restriction diagnosis. Keep invalid and missing tests visible in QualityAudit.

## SPC4 Check capability

Using fictional test specifications 58.9–65.1 L/h, calculate baseline Cp and Cpk from the stated within-process estimator. State assumptions: stable baseline, adequate measurement, suitable distribution and independence. Do not use a good Cp to claim the later monitoring period is stable. Review potential shift/measurement effects and compare with Worked-Reference.xlsx.

## Acceptance evidence

Keep the workbook formulas, chart, source checks, corrections and a concise interpretation. Use Worked-Reference.xlsx for comparison after your attempt. Tool access and actual Copilot edits must be recorded honestly.

# Lab 02 Copilot in Excel: FMEA and risk prioritisation

100 minutes | Lab-02-FMEA

## 1. Open the shared case

Use your previous workbook or this lab's standalone Starter.xlsx. Read Evidence, FMEA_Input and ScoringNote. Reuse the SPC interpretation if available; no completed Lab01 file is required.

## 2. Create the review table

Select FMEA_Input; submit FMEA1. Check all six IDs and columns. FM01 has observed low-delivery evidence but an unconfirmed restriction cause; FM04 is a prospective leakage scenario with no supplied leak event. Keep source status explicit.

## 3. Validate the ratings

Read the supplied provisional scale definitions. Higher D means poorer detectability. Every rating is a workshop judgement and must retain its rationale. Do not recalculate occurrence from 120 tests or treat a missing incident as proof of low risk.

## 4. Use formulas and charts

Submit FMEA2. RPNs: FM01=140; FM02=70; FM03=108; FM04=90; FM05=48; FM06=32. Sorted order is01,03,04,02,05,06. Highlight FM04 separately because S=9. Verify formulas remain attached to IDs after sorting and create an RPN column chart.

## 5. Run a short evidence-led debate

Submit FMEA3. Reliability examines mechanism evidence, operations examines practicality, safety examines consequence and approval gaps. Each challenges one claim; moderator retains unresolved gaps. These are simulated roles, not independent experts. The removal of a strainer is not effectiveness evidence for every failure mode.

## 6. Create an action view

Keep proposed actions separate from current controls. Add owner role, due-date proposal, required effectiveness evidence and status. Leave residual S/O/D blank or explicitly unassessed. Do not invent an implementation date or approved work instruction.

## 7. Save and peer review

Save FMEA_Review, priority chart and disagreement/evidence log. Exchange one row with a peer: identify function/mode/effect/cause/control and recompute its RPN. Explain why severity review can outrank the numerical ranking.

# FMEA Copilot prompts

## FMEA1 Structure the review

Use FMEA_Input and Evidence. Create FMEA_Review with function, mode, effect, potential cause, current control, evidence, supplied provisional S/O/D, RPN, action, owner and status. Keep the six mode IDs. Separate actual observations from prospective scenarios. Use ScoringNote for definitions and do not infer occurrence ratings from the short test series.

## FMEA2 Calculate and prioritise

Add an Excel formula RPN=S*O*D for every row. Sort a view by RPN descending while preserving mode IDs and source rows. Add a separate high-severity review flag for S>=9. Create an RPN column chart. Do not treat rank or low RPN as evidence of safety acceptance.

## FMEA3 Challenge with personas

Simulate reliability, operations and safety review lenses. First return one separate evidence-based concern per role. Run two concise challenge/response rounds, citing mode IDs and evidence records. A moderator returns agreed revisions, unresolved disagreement and missing evidence. Do not force consensus or invent approval. Keep residual ratings unassessed until action-effectiveness evidence exists.

## Acceptance evidence

Keep the workbook formulas, chart, source checks, corrections and a concise interpretation. Use Worked-Reference.xlsx for comparison after your attempt. Tool access and actual Copilot edits must be recorded honestly.

# Lab 03 Copilot in Excel: DMAIC and improvement reporting

90 minutes | Lab-03-DMAIC

## 1. Define the problem

Open WeeklyEvents and DefectCounts. These are an eight-week event register, separate from the120-test engineering investigation sample. Define the CTQ as acceptance deviations per logged test opportunity. It is not treatment efficacy or regulatory compliance.

## 2. Calculate exposure-aware rates

Submit DMAIC1. Sum baseline weeks1–6:93 deviations/1008 opportunities=9.22619%. Pilot weeks7–8:11/336=3.27381%. Label phase and denominator. A comparison with unequal observation periods should use rates, not raw count reduction.

## 3. Create the Pareto

Submit DMAIC2. Counts55,16,12,10 sum to93. Cumulative shares59.14%,76.34%,89.25%,100%. Use descending columns with a cumulative-percent line on a secondary axis. Categories prioritise investigation and do not establish root causes.

## 4. Interpret change honestly

Absolute reduction=5.95238 percentage points; relative reduction=64.51613%. The pilot has only two weeks, and strainer replacement plus verification occurred together. Do not call the result proof of one cause, permanent improvement or quantified financial savings.

## 5. Link Analyse to Improve

Use the SPC evidence and FMEA action list to propose a follow-up study: agreed CTQ, fixed test conditions, documented instrument check, comparable baseline/pilot scope and independent engineering review. Mark proposal status; do not prescribe hazardous operations.

## 6. Write the Control plan

Submit DMAIC3. Specify the metric, source, review frequency, responsible role, signal rule and escalation route under approved site procedures. Keep baseline limits fixed until a justified re-baseline is reviewed. Retain ACT-401-04 as open pending evidence.

## 7. Present one workbook summary

Prepare six cells: problem, baseline, pilot, evidence, limitations and proposed next decision. Save the rates, Pareto, DMAIC and ControlPlan sheets. Verify every management claim back to a source row and give a two-minute review to a partner.

# DMAIC Copilot prompts

## DMAIC1 Define and measure

Use WeeklyEvents, DefectCounts and Evidence. Draft a DMAIC table in Excel: phase, claim, metric, source, limitation, next action. Define the CTQ as delivery-test acceptance deviation, not chemical compliance. Calculate baseline and pilot rates using summed deviations divided by summed opportunities, not a simple average when denominators differ.

## DMAIC2 Analyse

Create a Pareto table and chart from baseline DefectCounts, including cumulative percentage. Reconcile the total with baseline WeeklyEvents. Compare the phase rates and express both percentage-point and relative change. Explain that fewer deviations after two concurrent changes do not isolate the cause or establish sustained improvement.

## DMAIC3 Improve and control

Draft a controlled follow-up trial and ControlPlan table: metric, source, frequency, owner, signal rule, review response and evidence. Link to SPC signals and FMEA actions. Preserve the open removed-strainer review. Do not claim causal proof, savings or operational approval not supported by the source. End with a six-cell management summary and unresolved questions.

## Acceptance evidence

Keep the workbook formulas, chart, source checks, corrections and a concise interpretation. Use Worked-Reference.xlsx for comparison after your attempt. Tool access and actual Copilot edits must be recorded honestly.